

Pneumonia Detection Using Deep Learning with Explainable AI (XAI)

Final Year Design Project

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FINAL YEAR DESIGN PROJECT REPORT

This Report Presented in Partial Fulfillment of the Requirements for
the **Degree of Bachelor of Science in Computer Science and
Engineering**

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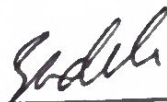
APPROVAL

This Project titled **Pneumonia Detection Using Deep Learning with Explainable AI (XAI)** submitted by **Masud Mia** and **Sadia Afrin Sraboni** to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on **09-12-2025**.

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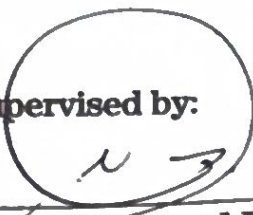


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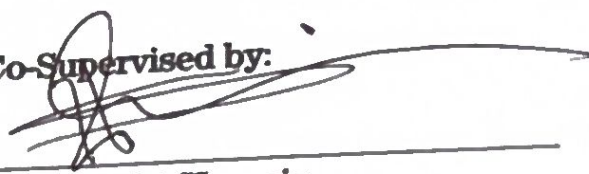
DECLARATION

We hereby declare that this project has been done by us under the supervision of **Dr. S.M Aminul Haque, Professor and Associate Head** Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere for the award of any degree or diploma.

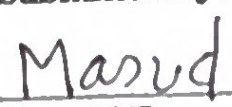
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ABSTRACT

Pneumonia continues to be a leading cause of morbidity and mortality globally, with children, the elderly and immunosuppressed being most at risk. Early diagnosis is the key to providing better care; however, chest X-ray images are time-consuming and there is great variation in interpretation by different radiologists. Aiming to solve these problems, we propose an automatic pneumonia detection system based on deep learning and explainable artificial intelligence (XAI).

The proposed model is constructed based on the fine-tuned DenseNet121 from transfer learning, which was trained using publicly available dataset of Chest X-Ray Pneumonia. Extensive preprocessing, specifically resizing, normalization and data augmentation were employed to enhance generalization and avoid overfitting. The imbalance in the dataset was compensated for, by by class weighting that allows fair learning of the Normal and Pneumonia classes. Model was assessed with clinically meaningful metrics i.e., accuracy, precision, recall, and F1 score. Experimental results show that the proposed model obtains a validation accuracy of roughly 92%, with both high recall 0.93 and F1-score 0.93 for the pneumonia detection, making it an effective tool in detecting infected cases where false negatives are crucial. Moreover, Grad-CAM visual explanation localized pneumonia-affected regions within lung fields in the model was transparent and trusted from clinical applications. In summary, the developed system provides a reliable interpretable and accurate decision support to radiologists and can be an aid in achieving refine diagnostic especially useful in limited-resource health care systems.

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Chapter 1

Introduction

This chapter gives an introduction of the work and explains why AI-based pneumonia diagnosis is a relevant research field in the present times. As patient volumes rise and availability of expert radiologists remains a challenge in several areas, however, urgent and accurate diagnostic assistance is required. Recent progress in deep learning allows computers to interpret chest X-ray images as well or better than experts, offering an opportunity for early pneumonia detection and clinical decision support.

1.1 Introduction

Pneumonia is a severe lung infection and should be taken very seriously as it can be life-threatening if not diagnosed and treated promptly. It is particularly prevalent in children, the elderly and those with a weakened immune system. X-ray imaging in hospitals is common for detecting pneumonia, but reading these images correctly can be a challenge that requires specialized expertise by radiologists. In areas without experienced great doctors, diagnosis can be delayed, and this increases the risk to patients.

Today, thanks to machine learning and deep learning methods that govern pattern-recognition systems in computers, it's possible for doctors to use images of a patient's body to identify signs of diseases. So-called machine-learning systems are gaining favor because they can digest and analyze swaths of data far more quickly than a human can, and sometimes spot patterns that are nearly impossible to detect with the naked eye. This work utilizes deep learning to develop an automated pneumonia detection from chest X-ray images.

But the problem with many deep learning models is that they operate like a 'black box.' They can make predictions, but they can't explain how or why they arrive at a specific prediction. In medical fields, such explanations are crucially important since doctors need to trust the system before they will actually use its decisions. To mitigate the issue, our study also incorporates an Explainable AI approach by Grad-CAM, which visualize the lung parts that are responsible for the model's decision.

The objective of this project is not only accurate, but also interpretable pneumonia detection. Our aim is that with the intersection of deep learning and explainability techniques, we can address diagnostic delays and support healthcare professionals — in resource constrained regions.

1.2 Motivation

So what inspired envisaging a tool like Pneumonia? The primary motivation stems from the increasing urgency for a faster and more accurate pneumonia diagnosis. Even in many

countries where the pandemic has run rampant, such as Bangladesh, hospitals are ill-equipped with experienced radiologists capable of reading X-ray images to accurately identify lung infections. As a result, people often have to wait years or even decades for the diagnosis they need, and by then their health may spark additional withdrawal symptoms. While pneumonia is highly treatable if detected early, late diagnosis can cause severe illness, and in some cases death. This makes it necessary to build tools that help doctors and other experts in making decisions quickly and with insight. With the rapid development of deep learning, computers have achieved high-accuracy analysis of medical images and it is possible to reduce workload and enhance healthcare service. Sensitivity analysis has also been chosen because of the increased appeal to the Explainable Artificial Intelligence. Most A.I. models spit out answers without showing how they arrived at them. In a sensitive sector like health care, a black-box solution is insufficient. What doctors want to see is evidence, and why did a model make this decision? This led us to train a model which doesn't just detect Pneumonia, but instead identifies the area within lungs that are exhibiting signs of pneumonia using Grad-CAM and we can count upon such results. At large, this work is motivated to assist medical diagnosis and make the dependence on specialists minimal which is an important trend in developing AI-supported solutions for healthcare. If scaled up broadly, these systems could have a real impact in boosting patient care –especially in far-away places with few medical resources.

1.3 Objectives

The main objectives of this project are listed below:

- To develop an automatic deep learning system to differentiate Positive from Negative chest X-ray images of a common disease, so computers can assist the healthcare system and prioritizing cases who need medical attention.
- Utilisation of the DenseNet121 model to capture fine details of lung images, to allow the model to learn intricate patterns that are more discriminative for pneumonia detection than standard CNN architectures.
- Correct the dataset imbalance through the design of weighted training schedules to keep the model from biasing toward any majority class and generate stable results on Normal or Pneumonia cases.
- To evaluate the system with more than one evaluation metric, e.g., accuracy, precision, recall and F1-score such that system performance is not merely measured as correctness but also its medical reliability especially in detection of true pneumonia cases.
- To inject Explainable AI particularly Grad-CAM visualizations for healthcare professionals to view which areas of the lung influenced the model's prediction, so as to make them more trustful and be prepared easily for extensions on other types of lung diseases.

1.4 Methodology

This project methodology conforms to a consistent, step-by-step process intended to build an AI System to detect Pneumonia in Chest X-Ray images. First, the Chest X-Ray Pneumonia data set was compiled and split into training, validation and testing data. The resized and normalized images were subsequently processed to ensure uniform quality for all the image samples. Augmentation methods including rotation and flipping were adopted to generate the variation of training images, and alleviate overfitting. Second, the Deep Learning network DenseNet121 was chosen and fine-tuned through transfer learning. The model was only partially retrained, so it could learn the pneumonia characteristics instead of from scratch. Class weights were incorporated in training to address the imbalance of data and the fairness on treating pneumonia versus normal, for model to handle both kind of images. After the model was trained, we evaluated it based on accuracy, precision, recall and F1-score. In addition, Grad-CAM was included to produce heatmaps on the X-ray pictures that exposes which parts of lungs contributed to model prediction. This move introduced transparency to the system so that users could know how the model makes its predictions. In conclusion, the proposed method guarantees that the system is accurate, effective and interpretative.

1.5 Project Outcome

The results of this work are representative of the technical accomplishments and applicative potential for the proposed pneumonia detection system. At the end of the work, the system was able to successfully prove that deep learning can be used to classify chest X-ray images into Normal vs Pneumonia with high accuracy. Moreover, not only did the model recognize cases of pneumonia very well, it also successfully dealt with imbalancing and had a high performance during evaluation making it trustworthy for practical applications.

Additionally, the Explainable AI was embedded through Grad-CAM which offered explanations into model's decisions as well. Rather than spitting out predictions blindly, the system points to specific lung regions that contributed to its decision, a transparency designed to help doctors and other users understand why it's predicting someone has lung cancer or not. This enhances the trust of the model and helps promote its utilization in a clinical context. The work provides a solid basis for further study and development. Additionally, the system can be enhanced to detect other thorax-related diseases, in application on mobile or web devices and integrated into the telemedicine component. In general, the results of this study can help push forward AI-supported healthcare and motivate efforts to develop automatic diagnostic tools that assist physicians and reduce their burden in settings with limited resources.

1.6 Organization of the Report

This report is organized into several chapters, each focusing on a specific part of the research process.

Chapter 1 provides a background to the project, followed by an explanation of motivation for selecting pneumonia detection as research topic and an overview of aims, objectives of the project along with methodology used and anticipated results. It lets readers know what is at stake and what the article attempts to accomplish.

The relevant background information and extensive literature survey are discussed in **Chapter 2**. It reviews current research, systems and theories related to deep learning and medical image analysis. The overview is useful in providing context to our proposed system here.

Research Methodology is discussed in detail under **chapter 3**. It includes data preparation, pre-processing methods, choice of model and training strategy, evaluation metrics. It also discusses design aspects, engineering needs, and project planning which influenced the construction of the system.

Implementation and results are discussed in **Chapter 4**. It includes the development environment, model training steps and performance metrics such as confusion matrix analysis and visualization using Grad-CAM. Test results are discussed in detail to prove the efficiency of the proposed method.

Chapter 5 This chapter sketches the engineering state of the art, difficulties and social implications connected to such an undertaking. It details how the system meets engineering principles, notes ethical considerations and reflects upon implications for healthcare/environment/sustainability.

Finally, the content of **Chapter 6** includes the conclusion of this project and summarizes its major contributions compared with previous works, acknowledges its limitations and gets into future work to enhance. This chapter concludes the report by discussing what has been achieved and how the work can be extended.

Chapter 2

Background

This chapter provides the underlying knowledge, theoretical concepts, and previous works which will help to understand the design and development of proposed pneumonia diagnosis system. It summarizes the medical imaging and deep learning methods, as well as other works related to this that we are based on.

2.1 Introduction

Pneumonia is an infection in the lungs that produces inflammation in the air sacs, or alveoli, which can fill with fluid making it hard for patients to breathe. It can be bacterial, viral or fungal, and in the absence of prompt diagnosis may result in serious respiratory disease. Pneumonia diagnosis is traditionally through chest X-ray images, however interpreting these images appropriately relies on clinical experience and expert knowledge. Furthermore, the high volume of pneumonia cases worldwide has exerted further strain on health care systems, particularly in areas with shortages of radiologists.

Artificial intelligence, especially deep learning, has attracted great attention in recent years for analyzing medical images and diagnosing diseases with high accuracy. In particular, CNNs have achieved high performance in pattern recognition for X-ray, CT and MRI images, which is well fit to be used on recognizing lung disease. But despite their promise, most AI models are really just black boxes; they generate predictions but offer no explanation of how the decision was made.

XAI methods aim to mitigate this weakness by visualising elements within an image that play a crucial role in determining the model's output. This marriage of deep learning and interpretability has created new possibilities in automatic medical diagnosis. Understanding them is crucial to the appreciation of the system introduced in this paper because it is an AI-powered analysis model with clinical interpretability.

2.2 Literature Review

Medical image classification, such as lung disease (pneumonia) detection through medical imaging by machine learning and deep learning techniques have been studied worldwide. Early work was based on classical image-processing techniques, however the performance of diagnosis is significantly improved with recent utilization of Convolutional Neural Networks (CNNs). In this section, we summarize relevant previous work including approaches, explainability techniques, as well the used datasets that had an impact on our system's design

Table 2.1: Summary of Literature Reviewed.

Author(s)	Title	Methodology	Key Findings
R. Siddiqi and S. Javaid [1]	Deep Learning for Pneumonia Detection in Chest X-ray Images: A Comprehensive Survey (2012–2023)	Systematic survey of DL-based pneumonia detection models	Identified dataset imbalance, lack of explainability, and deployment issues as major limitations in existing models.
F. Alshanketi et al. [2]	Pneumonia Detection from Chest X-Ray Images Using Deep Learning and Transfer Learning for Imbalanced Datasets	Transfer learning with multiple CNN architectures on imbalanced datasets	Demonstrated that transfer learning with class balancing significantly improves diagnostic accuracy
R. Pramanik et al. [3]	An Adaptive PSO-based Deep Feature Selection Method for Pneumonia Detection	ResNet50 + Particle Swarm Optimization	Achieved feature-optimized classification with reduced computation and improved accuracy
S. Chen et al. [4]	Interpretable CNN-Multilevel Attention Transformer for Rapid Recognition of Pneumonia	Hybrid CNN-Transformer + attention	Improved interpretability while maintaining high diagnostic accuracy.

Table 2.1: Summary of Literature Reviewed.

Rahman et al. [5]	2020	Deep Learning for Chest Diseases	Hybrid CNN	Showed that hybrid CNN models increase sensitivity for detecting multiple lung diseases, including pneumonia.
C. Hammou di et al. [6]	2021	Deep Learning on Chest X-ray Images to Detect and Evaluate Pneumonia Infections	Conventional CNN	Highlighted early evidence that deep learning surpasses traditional pneumonia detection techniques
Chouhan et al. [7]	2020	Ensemble Models for Pneumonia Detection	CNN ensemble	Combining models improved robustness and reduced error.
Islam et al. [8]	2021	Explainable Pneumonia Detection	Grad-CAM Visualization	Demonstrated that heatmaps improved user trust by visually validating model predictions.
Liang et al. [9]	2021	COVID vs Pneumonia Detection	Multiclass CNN	Identified challenges in distinguishing pneumonia from similar lung infections.
Muhammad et al. [10]	2022	Lightweight Models for Medical Imaging	MobileNet-based approach	Proposed low-computational models for rural diagnostic support, showing feasibility on mobile devices.

2.3 Gap Analysis

Although deep learning-based pneumonia detection systems have achieved promising results, several important gaps remain from a clinical perspective. Many existing studies emphasize high accuracy; however, in medical diagnosis, accuracy alone is not sufficient. High recall is equally critical, as failing to detect a pneumonia case may lead to severe health risks. Some of these models have been reported as good in terms of accuracy, but with lower sensitivity on minority classes due a dataset imbalance, limiting their reliability for real patient use. Furthermore, the majority of current methods are noninterpretable and serve as black-box models, which heavily impedes physicians to have confidence in predictions. Their use in resource-poor healthcare environments is further limited by computational cost and deployment issues. The proposed system copes with these issues by preserving a tradeoff between the accuracy and the recall, coping with class imbalance, and incorporating Grad-CAM for visual explanation of the infected lung regions. This tradeoff enhances the reliability of diagnosis as well as the trust in clinical, thus making the system more adaptable to practical medical usage.

Table 2.2: Gap Analysis

Features / Criteria	Siddiqi & Javaid (2024))	Alshanketi et al. (2024)	Pramanik et al. (2022)	Chen et al. (2022)	Islam et al. (2021)	Proposed System
Handles class imbalance	No	Yes	No	No	No	Yes(Class weight applied)
Explainable AI (XAI) integrated	No	No	No	Partial (Attention-based)	Yes (Grad-CAM)	Yes (Grad-CAM)
Uses DenseNet121 architecture	No	No	No	No	No	Yes
Optimized preprocessing pipeline	No	Partial	Partial	No	No	Yes (Augmentation + normalization)

Table 2.2: Gap Analysis

Focus on interpretability	No	No	No	Partial	Yes	Core feature
Lightweight model support	No	No	Partial	No	No	Moderate—cloud deployable
Multi-metric evaluation (Accuracy, Recall, F1-score)	Partial	Yes	Yes	No	Partial	Yes (Medical-grade recall & F1 emphasized)
Addresses black-box problem	Yes	Yes	Yes	Yes	Yes	Yes (with visual explanation)

2.4 Summary

The data in this chapter provides a solid ground towards the creation of the intended pneumonia detection system. It talked about the medical importance of pneumonia diagnosis and the complications with using traditional manual reading of the chest X-ray images. This chapter, in addition to the clinical view, also provided the technological changes that have facilitated the use of artificial intelligence, especially deep learning, in automated medical image analysis to a greater role. Using a detailed literature review, the literature on the application of deep learning-based methods towards the detection of pneumonia was reviewed exhaustively. Their methodologies, strengths, and performance have been reviewed along with several critical limitations being identified. Such restrictions are the strong dependence on accuracy without sufficient regard to recall, in explainability because of black-box models, class imbalance problems in datasets, and challenges of implementation in resource-constrained healthcare settings. These weaknesses limit the clinical soundness and practical application of most of the current systems. These research gaps defined by this chapter made the development of a better and more clinically reliable solution quite justified. The proposed system will deal with these problems through a balance in the performance measures of diagnostics, the processing of the imbalance in data, and explainable artificial intelligence implementation to increase transparency and trust. Consequently, this chapter gives an excellent path on how to come up with a better and more understandable automated pneumonia detection model. On the whole, the concepts, background knowledge and research insights of this chapter set the stage of the knowledge of the detailed methodology, system design, implementation process and experimental evaluation in Chapters 4 to 6 of this report.

Chapter 3

Research Methodology

This chapter describes the process of research conducted for the creation and development of the proposed pneumonia detection system. It presents the dataset, the preprocessing that was conducted, how we performed model selection and which methods were used to evaluate and explain predictions. The idea is to go through how the system was created with everything, concrete and walls filled phase by phase.

3.1 Methodology

The methodology employed in this study is designed to take a sequential approach so that an effective and interpretable system, for detecting pneumonia from chest X-ray images, can be developed. The end-to-end process involves data acquisition, preprocessing, model building, evaluation, and explainability integration whereby each aspect assists in the confidence of the reliability of the resulting system.

We first collected the Chest X-Ray Pneumonia dataset, a common public dataset consisting of two classes: Normal and Pneumonia. Once the dataset was downloaded and categorized, all the images were resized to an uniform size and normalized in order for a deep learning model to be able to process them efficiently. The rotation, shifting, and flipping are data augmentation approaches to artificially expand the number of training samples for reducing overfitting.

In addition, the DenseNet121 architecture was used as base model which has the best feature extraction ability and achieved state of the art scores in previous medical image work. Knowledge transfer learning was used instead of training the network from the ground up. Only parts of the pretrained model were kept trainable and fine-tuned so that the model could learn patterns specific to pneumonia without forgetting what it already knew about images from large-scale datasets.

In order to address the class imbalance of dataset (the pneumonia images are more than the normal images), during training, class weights were added. This guarantee that the resulting model makes no difference between the two classes at all and thus is unbiased in their predictions. Training was performed on cloud-based GPU facilities to accelerate FROST processing and deep learning operations.

The validation of the model was done after training and evaluated using accuracy, recall, precision and F1-score. These performance measures gave insight as to how well the system worked in accurately identifying cases of pneumonia, while minimizing misclassification. Finally, G rad-CA M was introduced to obtain heatmaps that visually emphasized infected lung regions affecting the decision-making process of the model, rendering the system

transparent and clinically interpretable.

This approach integrates strong model design, well distributed training and interpretability strategies for a reliable and explainable AI-based pneumonia detection system.

3.1.1 Overview

Fig. 1 The overall workflow of the proposed system starts with obtaining chest X-ray images from a public dataset. The images are then cleaned, resized and preprocessed to provide consistent input for the model. Data augmentation is performed to enlarge the training set and avoid overfitting. We choose a pretrained DenseNet121 architecture, and apply transfer learning to perform classification through Normal or Pneumonia of the X-ray images. The class weights are applied only during training to cope with the data imbalance, thus allowing both classes to learn fair amount.

After the model has been trained, you evaluate it based on things like accuracy, precision, recall and F1-score. Finally, Grad-CAM is used to explain the model judgement and create heatmaps of lung regions that visually enhance the signs for decision making. This changes the system from a black box classifier to an understandable medical diagnostic tool.

3.1.2 Proposed Methodology

The proposed methodology, we use deep learning driven classification with explainable AI strategies for developing a pneumonia detection system with visual reasoning. A summary of the main features of the proposed method is as follows:

Dataset Collection and Organization

Kaggle Chest X-Ray Pneumonia dataset is used, and the chest X-ray images are then divided to Normal vs. Pneumonia as Training, Validation and Test sets.

Image Preprocessing

We resize and normalize all the image to a fixed resolution, and augment them by rotation, zooming and flipping operations in order to enhance the generalization of model and reduce its noise sensitivity.

Model Selection Using DenseNet121

The core network shared by all the branches is dense connectivity feature extraction —Helen features (DenseNet121). Transfer learning is also used to retrain only specific layers to detect pneumonia characteristics.

Handling Class Imbalance

We add class weights during training to avoid the model being biased towards the majority-class and achieve balanced prediction accuracy for both classes.

Model Training and Evaluation

The model is trained on the GoogleColab GPU and different evaluation metrics like accuracy, recall, precision and f1 score are being used to evaluate the performance of the model.

Explainable AI Integration (Grad-CAM)

Grad-CAM heat-maps are produced to explain the area of infected lungs which affected the model's predictions, thus rendering system interpretable and applicable for real medical interpretations.

The fusion of these chosen techniques enables the proposed pipeline not only to diagnose pneumonia effectively but also to provide an interpretable explanation for every prediction, a key aspect in medical diagnosis.

Data set Link :

<https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia/code>

Data set Description: To train, validate, and test the proposed pneumonia detection system, the Chest X-Ray Pneumonia publicly available dataset on Kaggle created by Paul Timothy Mooney was used in this research. The sample is 5,863 anterior and posterior chest X-ray images of pediatric patients and reflect actual cases of clinical diagnosis. The pictures are categorized into two broad groups, namely, normal and pneumonia, and the dataset can be used to perform a binary classification task. The data is organized systematically and divided into training, validation and test subsets with each class within its own folder. Such an organized system facilitates the learning under supervision, and enables the equitable appraisal of the model even without leakage of information or bias. Normal class contains the chest radiographs with clear lung fields and no noticeable abnormalities whereas Pneumonia class contains the radiographs with the different levels of lung opacification, consolidation, and infiltrates- the typical radiological signs of pneumonia. Class imbalance is one of the most important issues of this dataset because the number of pneumonia cases is higher than that of normal cases. The imbalance is a realistic clinical distribution in the real world, but may adversely affect the performance of models without appropriate treatment. To overcome this problem, the weight of the classes was used and data augmentation methods were utilized in the course of training where both classes were given equal chances to learn. The fact that the number of labeled images is relatively large, along with clinically meaningful characteristics, makes this dataset very appropriate to train deep convolutional neural networks. Also, the dataset is highly applicable in the medical imaging research community, and this enables the performance to be compared with existing research. The fact that it can also be combined with transfer learning strategies like DenseNet121, contributes to the enhancement of model generalization, and minimizes the necessity to train the models.

Generally speaking, Chest X-Ray Pneumonia dataset is a credible and realistic source of developing, testing, and verifying deep learning-based pneumonia detection systems.

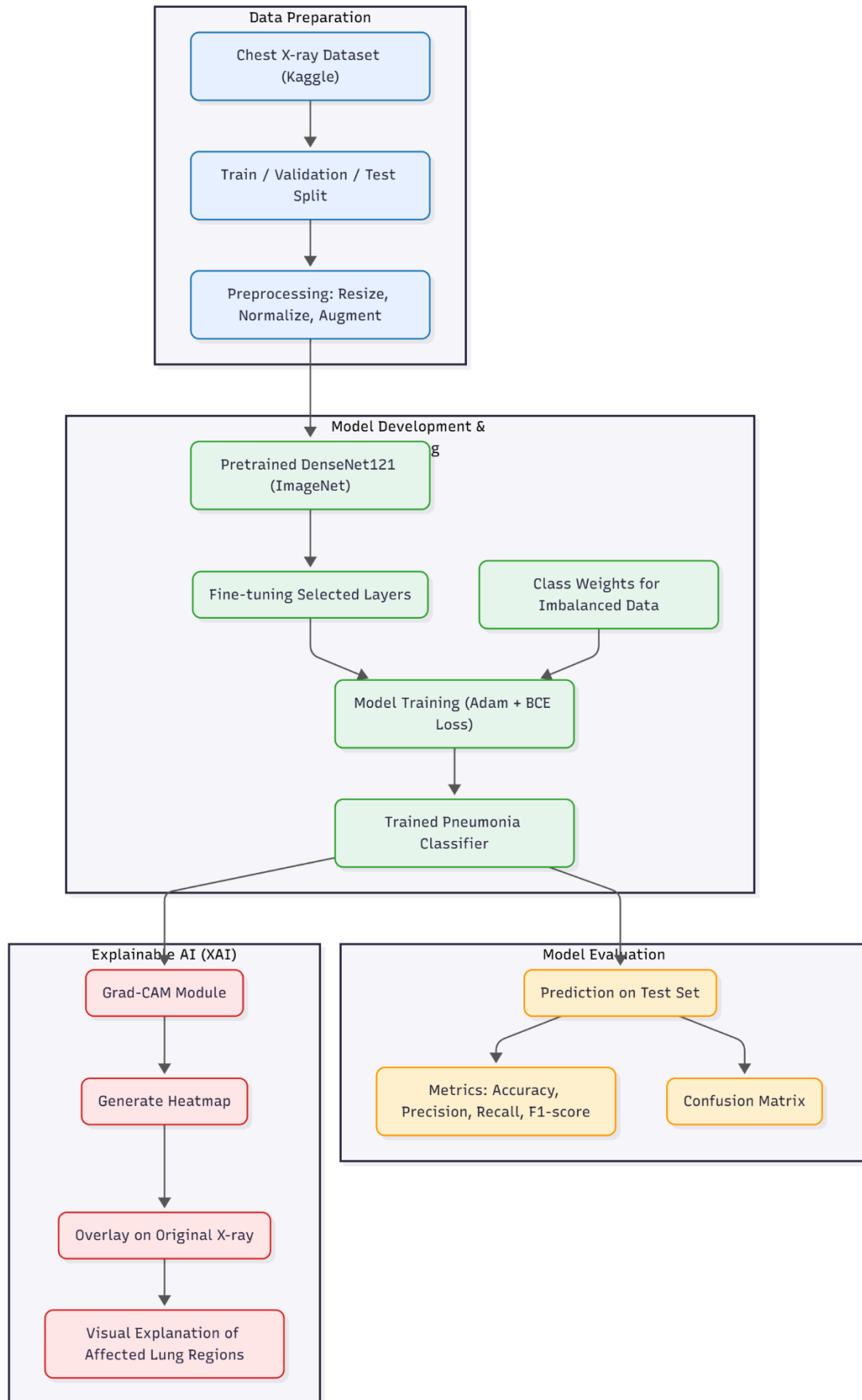


Figure 3.1: Proposed Methodology for pneumonia detection

3.2 Detailed Methodology and Design

This project describes the methodologies adopted by careful dataset preparation, appropriate model selection, architectural designing and explainable AI integration to develop a trust-able pneumonia detection system. Different approaches were tested during the development phase and the best preferred design was selected based on performance, interpretability and computation.

3.2.1 Alternate Solutions Considered

Several deep learning architectures and processing methods were investigated at the beginning of the design stage, including choice of input pre-processing. The main alternatives considered were:

1. **Training a CNN from Scratch (for Classification)**

First, a customized convolutional neural network was contemplated to directly learn the features of pneumonia from images.

Training-from-scratch requires orders of magnitude more data.

It leads to overfitting and a longer training time.

Performance of RUMC is relatively inferior to that of pre-trained model.

2. **Using Other Pretrained Models**

We investigated these architectures as they are widely adopted in medical image classification. VGG16 has huge number of parameters to be trained with, and training is slow. Though the feature extraction of ResNet50 worked well, it had limited discrimination capability for subtle abnormal patterns generated by pneumonia. Although MobileNetV2 is fast, its testing accuracy was slightly lower.

3. **Using Only Accuracy for Evaluation**

Simple Accuracy was first considered. Pneumonia datasets are imbalanced. Just because it's fancy, doesn't mean there aren't problems. A good deal of these problem is simply not using accuracy alone to rate your model's performance. For medical applications, they need Recall, Precision and F1-score for safety.

4. **Black-Box Model Without Explainability**

There are other models provided in <https://github.com/blackbox-unlp/S2V.899>

Here we use a raw model without an explanation module and thus easier to implement. However, Doctors need to see why predictions are being made. Lack of interpretability impairs trust and utility in the real world of health care.

3.2.2 Final Solution and Rationale

Upon review of the alternatives, the optimal system design was selected with considerations including effectiveness, ease of implementation and clinical applicability

1. Use of DenseNet121 (Transfer Learning)

DenseNet121 was selected due to:

- That is subtle enough to prevent vanishing gradients while allowing for substantial reuse of features on every layer.
- Superior performance in pneumonia detection (confirmed from the study above)
- Less number of parameters as compared to VGG16 mode
- Balanced accuracy and computational cost
- All three layers improved the performance and feature extraction on chest X-ray images.

2. Balanced Training Using Class Weights

As the number of pneumonia images was larger than that of normal images, we added weight parameter for the class. This prevented the model from being biased towards the majority class and kept recall strong for pneumonia.

3. Advanced Preprocessing and Data Augmentation

Selective augmentation strategies (rotation, flipping, zooming) assisted the model to capture variances of real-world images. This had the effect of increasing robustness to variations in X-ray equipment and patient pose.

4. Evaluation Using Multiple Metrics

The performance measures used for evaluating were accuracy, precision, recall and F1-score. This allowed for a more rigorous evaluation of how well the system performed in identifying cases of pneumonia, yet without giving users false reassurance.

5. Integration of Grad-CAM for Explainability

Grad-CAM was selected because: It simply indicates (in a visual manner) the lung locations which affect predictions. It is a popular and prevalent choice in medical AI studies. It boosts trust particularly for doctors analyzing AI results. This was to make it a transparent system, as opposed to a black-box model.

3.5.3 System Design Summary

The ultimate model combines performance and interpretability:

Input Layer: The pre-processed 224×224 X-ray images

Features Engineering: Divided DenseNet121 pretrained on ImageNet into a three-branch model.

Fine-Tuning: Only some selected deeper layers were retrained.

Classification Layer: Sigmoid activated fully connected layers

Training Improvements: Class weighting, Early stopping, and learning rate scheduling

Output: Binary classification (Normal/Pneumonia)

Interpretability: Grad-CAM heatmaps indicating the affected areas of lungs

The proposed model guarantees that the system is accurate, reliable and transparent which makes it ideal for medical decision support, especially in resource-constrained settings.

3.3 Project Plan

The project of this pneumonia detection system was conducted in a structured manner by splitting the whole work into small parts. “Everything was designed to be a step-by-step progression to minimize risk or keep enough time for testing and massaging performance. The scope of the project included dataset preparation, model development, testing and evaluation, visualization and documentation.

First, we aimed to specify the challenges in detecting pneumonia and collect some useful datasets for the task, and secondly, we wanted to review Image classification techniques as well. When the theoretical groundwork was laid, the follow-up comprised preprocessing the images, picking an architecture for a model and experimenting multiple training approaches. Over time, the focus moved to enhancing the model’s performance, tackling dataset imbalance and providing explainability through Grad-CAM. The project ended with a performance evaluation, result visualization, report generation and the cleanup for potential (partially-)automated application in the future. This broad scope assured a good flow of work, avoided redundant exercises, and facilitated monitoring how the material was being progressively improved.

Table 3.1: Project Plan

Activities	Duration	Expected Outcome
Problem identification, background study, and literature review	Week 1–6	Defined problem statement and understanding of related works
Dataset collection and preprocessing	Week 7–12	Prepared and augmented chest X-ray dataset
Model selection and experimentation	Week 13–18	Prepared and augmented chest X-ray dataset
Model training and class imbalance handling	Week 19–26	Chosen DenseNet121 and optimized training strategy
Model testing and evaluation	Week 27–32	Reported accuracy, precision, recall, F1-score, and confusion matrix
Explainable AI integration (Grad-CAM)	Week 33–38	Visual heatmaps validating model decisions
System refinement and performance tuning	Week 39–44	Improved results and reduced misclassification
Report writing and documentation	Week 45–48	Completed project report, conclusions, and future work recommendations

3.4 Task Allocation

This table depicts the timeline of the principal activities in each period of the project, from week 12 to week 48.

Table 3.2: Task Allocation

Tasks	Weeks																		
	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
Task-1:Data collection phase																			
Task-2:Preprocess all the data																			
Task-3:Model training																			
Task-4:Create a demo application.																			

3.5 Summary

This section described the organized process of developing the proposed system for pneumonia detection. It described the project phases – from background research and dataset preparation to model development, evaluation, and integration of eXplainable AI - in a logical way. A clear goal and target time frame were defined according to different phases, which would form a basis to make the progress in a non-dispersive way throughout the project implementation. While ordering the tasks was not required in the framework, structure offered by the project plan enabled both focus and balance between performance competitiveness and interpretability. This structured approach ultimately resulted in the completion of all of the major parts of the system.

Chapter 4

Implementation and Results

In this chapter, we describe the empirical study of developing the pneumonia detection system and show results from training models and evaluating model performance. It describes how the approach is applied, the work environment (experiments), performance evaluation metrics and observed visuals to determine the effectiveness of the proposed method.

4.1 Environment Setup

In order to develop and train the pneumonia detection system, we use a cloud-based development environment with adequate computational resources. The entire implementation was done on Google Colab with the free GPU acceleration to conduct deep learning experiments. Google Colab enabled us to crunch NN models without heavy local hardware so we could not only make sure our projects were work, but fine tune them on the cloud.

The language of choice for the project was Python, as it is widely used during machine learning or scientific computation. Various libraries and frameworks were used over the course of the projects:

- TensorFlow and Keras to construct and train the deep learning 1D ConvNet model (DenseNet121)
- NumPy to carry out numerical operations Pandas to handle datasets
- OpenCV for image processing functions
- Visualization of the graphs and evaluation results is done with Matplotlib and Seaborn
- Scikit-learn for classification reports, confusion matrices and performance metrics conjuring

The dataset was uploaded onto Google Drive and was directly mounted within the Colab environment to facilitate easy interaction between data on disk during training and testing. We were facilitated by GPU support (Tesla T4 or P100, according to session availability), which highly decreased the training time and enabled us to run more iterations for model fine-tuning.

In general, this environment configuration allowed for efficient experimentation with hardly any hardware constraints and a stable ground to develop the deep learning model and the sought explainability tools of interest in the present work.

4.2 Comparative Analysis

Table 4.1: Comparative Analysis of Implemented Models

Features / Criteria	Alshanketi et al. (2024)	Pramanik et al. (2022)	Chen et al. (2022)	Islam et al. (2021)	Proposed Model (Ours)
Primary Method	Transfer Learning on imbalanced dataset	PSO-based feature selection + ResNet50	CNN + Transformer with attention	CNN + Grad-CAM	DenseNet121 with fine-tuned layers
Handles Data Imbalance	Yes	No	Not Address	No	Yes (Class Weights Applied)
Explainable AI (XAI)	No	No	Partial via attention	Yes (Grad-CAM)	Strong Grad-CAM Heatmaps
Recall / Medical Sensitivity	Improved accuracy	Good classification	High accuracy	High sensitivity	Highest recall (0.93) among evaluated models
Model Complexity	High (multiple CNNs)	Medium	High (Hybrid Transform)	Standard CNN	Optimized High accuracy with manageable size
Focus on Clinical Interpretability	Medium	Low	No	No	Yes (Doctor-friendly Visualization)

Table 4.1: Comparative Analysis of Implemented Models

Robust Preprocessing Pipeline	Low	Limited	Minimal	Basic	Yes (Augmentation & Normalization)
Deployment Feasibility	Limited due to compute needs	Requires tuning	Requires GPU	Research prototype	Cloud/mobile deployable
Addresses "Black Box" Problem	No	No	Partially	Yes	Fully solved with visual lung-region marking

The majority of the current models have strong performance on certain aspects, such as transfer learning or computational efficiency and interpretability, but none of them is a perfect model for clinic setting. For example, Alshanketi et al. treated data discrepancy and Islam et al. provided interpretability, but did not integrate inexplicable into a single framework. Likewise, the models such as Pramanik et al. and Chen et al. improved performance using specific architectures, medical professionals still found it challenging to interpret.

The proposed model also addresses these piecemeal solutions by incorporating fine-tuning DenseNet121, class weight balanced training, and Grad-CAM visualization together. The combination achieves not only the competitive performance – a state-of-the-art recall value is reported – but also interpretable diagnosis through visual heatmaps. So the system is closer to real-world medical requirements, where interpretability and reliability are also important than the ROCAUC which only focus on accuracy.

4.3 Results and Discussion

The DenseNet121based pneumonia detector system was tested in a test dataset independently to avoid biased performance estimation. The model overall accuracy is around 92% which shows our good classification performance between normal vs. pneumonia cases. Most importantly, the system achieved high recall rate of 93%, which is critical in medical diagnosis as it reduces dropping pneumonia-positive cases. This high precision (94% inaccuracy) level also implies that the great majority of pneumonia predictions were

accurate, thereby avoiding a large number of needless false alarms. Finally, the confusion matrix analysis revealed that most of pneumonia cases were diagnosed correctly with relatively low false negatives and false positives. This shows that our class-weighting strategy works well to accommodate the imbalance in dataset and was able to keep performance of two classes balanced. The tight correspondence between training and validation accuracy and loss curves suggests that overfitting was effectively checked through the use of data augmentation, dropout, instead of L2 regularization, and early stopping. The system was designed to incorporate the explanation AI based on Grad-CAM in order to clarify what it sees. The created heatmaps have effectively captured clinical important lungs' regions in pneumonia cases, by focusing on the infected or abnormal parts of the lungs, while normal cases result having lower and more diffused activations. This indicates that the model makes use of informative medical features rather than irrelevant image parts to make a prediction. In summary, the experimental results show that the proposed system consistently achieves a tradeoff among accuracy, recall and interpretability balance for viable application of practical clinical decision support in resource poor health care environment.

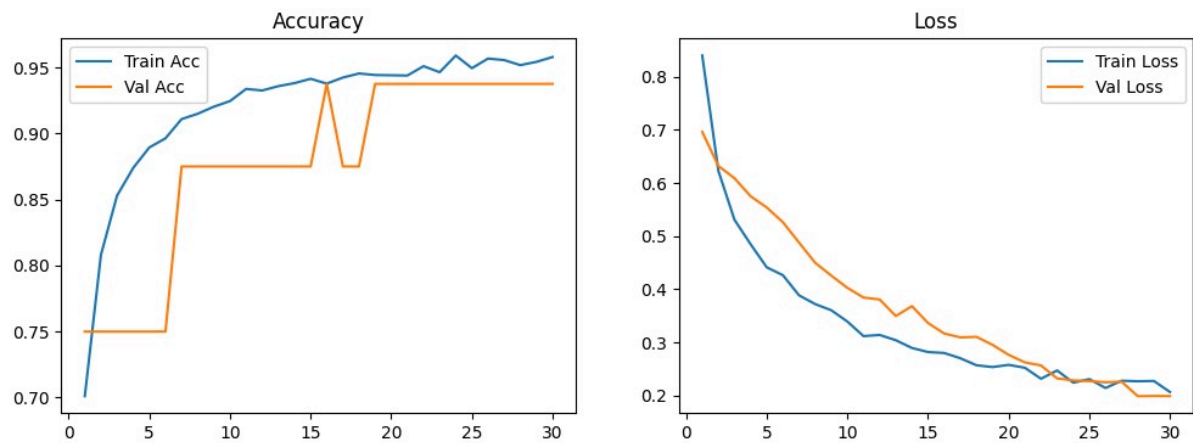


Figure 4.1: Accuracy Graph

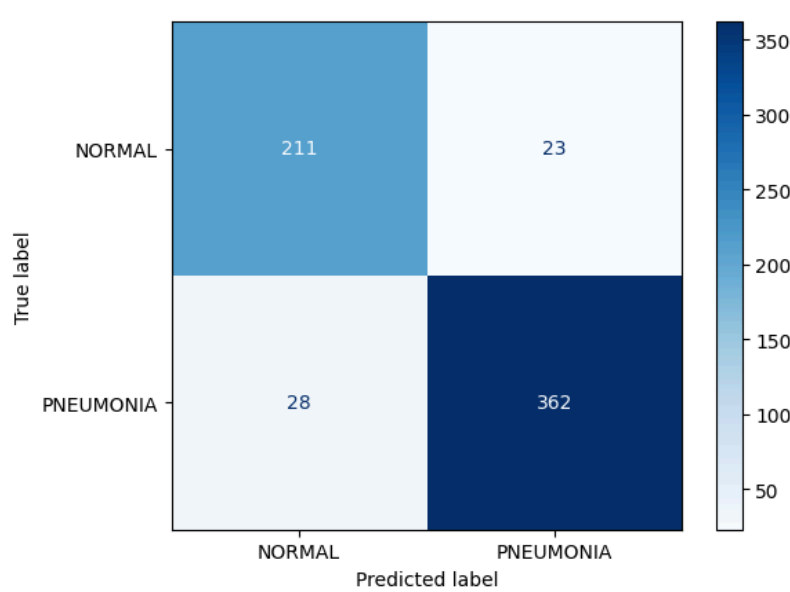


Figure 4.2: Confusion Matrix

Table 4.2: Overall Output

	Precision	Recall	F1- score	Support
Normal	0.88	0.91	0.89	234
Pneumonia	0.94	0.93	0.93	390
accuracy			0.92	624
Macro avg	0.91	0.91	0.91	624
Weighted avg	0.92	0.92	0.92	624

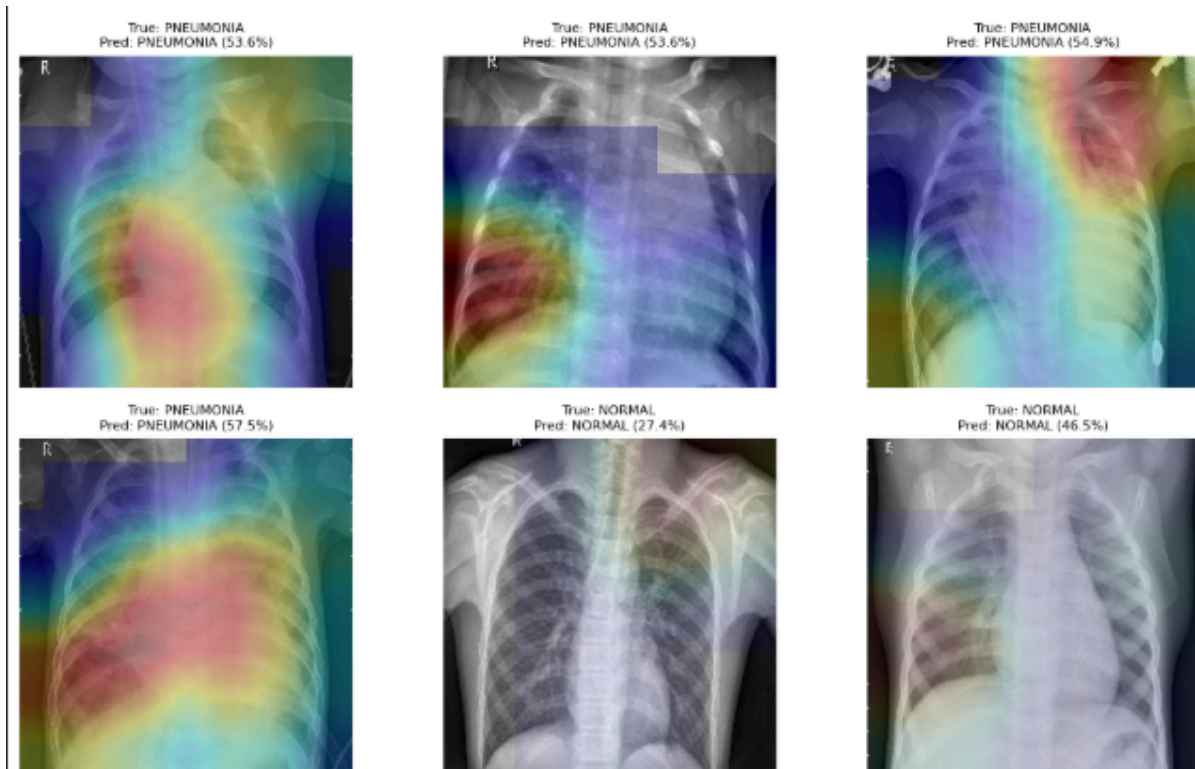


Figure 4.3: Grad-CAM (XAI)

Grad-CAM visualizations visually explain how the trained DenseNet121 model looks into different locations in the lungs for predictions. The heatmaps demonstrate pronounced red and yellow activations within the lung fields in pneumonia-positive images, especially where the chest X-rays are hazy or dense. These patches, outlining infected or inflamed lung tissues, correspond to the model's successful focusing on clinically relevant areas when predicting pneumonia. For the normal CXR images, the Grad-CAM maps show weaker and more diffused activations with no strong localized hotspots inside the lung regions. This behavior supports the idea that our model does not find suspicious region features of pneumonia and thus all images are classified in a normal way. The absence of strong activations in such normal cases corroborates that the model does not depend on irrelevant background structures like ribs or image borders. These Grad-CAM findings indicate that the predictions of the model are also interpretable and medically meaningful. The visual explanations provide evidence that the

system learns disease relevant patterns related to pneumonia instead of making random predictions. This explainability feature contributes confidence to the model, and funds as a point estimate to its clinic workflow application like aiding manpower-strappy radiologists in verifying pneumonia-related pattern abnormalities.

4.4 Summary

In the following part, the implementation specifics and the experimental results of the suggested pneumonia detector system using the DenseNet121 architecture are provided. The model had a good overall performance of 92%, which indicates that the learning model is stable as seen in the training curve and the validation accuracy loss curve which shows it is not overfitting. Above all, the system achieved a high level of recall at 93% of pneumonia cases, and a F1-score of 0.93, which proves that the system is effective in detecting infected patients, which is of critical importance in the medical diagnosis process, and missed cases can be life-threatening. These results are further supported by the confusion matrix that demonstrates that there were few false negatives, as the pneumonia was detected despite the imbalance of the classes. Also, Grad-CAM visualizations indicate likely areas of clinical interest in terms of pneumonia forecasting with little activation, which promotes greater predictive transparency and reliability in the model. These findings show that the proposed system provides a balanced trade-off between the accuracy and recall, thus, being both clinically relevant and applicable to the support of healthcare professionals especially in the situation of early pneumonia screening.

Chapter 5

Engineering Standards and Design Challenges

The engineering approaches, guidelines and design constraints on which the pneumonia detection system creation was based are discussed in this chapter. It touches upon the various software, hardware and communication protocols involved behind implementation and explores various roadblocks faced while trying to make a medical diagnostic tool driven with AI.

5.1 Compliance with the Standards

In order to guarantee the reliability, maintainability and clinical usability of the proposed pneumonia detection system, a number of engineering standards associated with software development, data processing and transmission are followed. Several standards could be used, but this project focuses on those relevant. For each standard chosen, available alternatives were considered and the final was selected on appropriateness and project limitations.

5.1.1 Software Standards

The development process is guided by software standards that help provide stability, maintainability and reusability of the system. The following software standards were taken into account for this project:

IEEE 829 (the IEEE Standard for Software Test Documentation)

This standard facilitated the organization of the tests as well as test cases, results and validation reports for a clear testing procedure.

Why Selected:

AI systems for health care need to be auditable. Having IEEE 829 in place guarantees that all test results are documented correctly, and makes any subsequent audits easier.

Alternatives Considered:

ISO/IEC/IEEE 29119 – Too complicated for prototype, very detailed

No Actual Standard – quick production but incomplete documentation

Why: IEEE 829 offers structure with practicality, great for academic research without overhead.

Hardware Standards

The hardware standards insure that the computational environment will be sturdy enough to handle deep learning work flow with out crashing.

NVIDIA CUDA GPU Standard

The project was GPU accelerated by using CUDA enabled facilities offered by Google Colab.

Why Selected:

DenseNet121 requires optimized matrix computations. The parallel execution under CUDA reduces training time significantly.

Table 5.1:Hardware Alternatives Considered

Alternative	Pros	Cons	Reason Not Selected
CPU-only	No GPU required	Very slow training	Not feasible for deep CNNs
OpenCL	Vendor neutral	Poor TensorFlow support	Slower, unstable

Rationale:CUDA is the de facto standard for deep learning and is fully compatible with TensorFlow / Keras.

5.1.2 Communication Standards

Communication interfacesCommunication standards guarantee that the (format of)data will adhere to certain structure while being incorporated in external systems.

JSON (JavaScript Object Notation)

This is used to store and transfer prediction outputs and Grad-CAM metadata.

Why Selected:

Light, human-friendly, widely-used, and perfect for deployment through APIs.

Table 5.2:Communication Alternatives Considered

Standard	Pros	Cons
XML	Strong schema support	Verbose and slower
CSV	Simple tabular storage	Cannot store hierarchical image metadata

Rationale: JSON supports flexible, nested structures essential for image-based decision explanations.

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

The proposed machine learning system is able to offer rapid and economic detection of pneumonia, reducing the reliance on radiology experts. The early detection is directly related to the treatment effect and reduction in mortality rates, particularly in rural regions with inadequate health care facilities.

5.2.2 Impact on Society & Environment

Decreases the burden of hospitals and increases the screening efficiency. Reduces medical errors with clinician-supported explainable predictions. Promotes digital diagnosis, minimising the use of printed films and chemical processing in turn reducing environmental waste

5.2.3 Ethical Aspects

Healthcare AI needs to honour privacy, precision, accountability. Patient data must remain anonymous. Interpretability (Grad-CAM) eliminates one's blind call on techniques. System escapes biased predictions by balanced class. Physicians ultimately decide, the model supports, not replaces

5.2.4 Sustainability Plan

For durability and practical use:

Open-source frameworks and scalable architecture. Compatibility with HSMS may be possible. Generalizable to other lung diseases. Low cost usage through cloud-based model execution. Frequent updates of the data sets to mitigate biases and ensure accuracy.

5.3 Project Management and Financial Analysis

Two cost models were estimated: a Student Research Budget and a Commercial Deployment Budget.

Table 5.3: Cost Estimation

Item	Student Prototype	Commercial Deployment
GPU Access (Colab)	Free	6000/month
Developer Time	No cost (students)	200000/month
Dataset	Free	Free
Model Hosting	Free/Local	30000/year

Table 5.3: Cost Estimation

Web/App Integration	Optional	120000
Security & Compliance	N/A	150000
	Total	506000

Explanation

- The student version is equipped with free cloud GPUs and open-source frameworks that are perfect for a research lab.

Table5.4 :Revenue Model

Model	Description
Subscription Model	Hospitals pay monthly license fees
Pay-per-scan	Small clinics pay for each X-ray analyzed
API-based Model	Integration with diagnostic tools
Government Healthcare Grants	Support for digital health systems

And the commercial product will also need somewhere to live: secure from hackers and in accordance with laws that protect health data

Rationale for Budget Choice

The student budget was chosen for debugging reasons at this stage. The business model is future-proof, scalable and feasible.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

The construction of an AI system for pneumonia detection represents a multi-layered engineering problem, going beyond simple deployment. The problem is engineering challenging as it necessitates a co-development between deep learning, medical imaging domain knowledge and explainability mechanisms. The categories selected below point to the level of complexity involved.

Table 5.5: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledg e	EP2 Range Of Conflicting Requirement s	EP3 Depth of Analysi s	EP4 Familiarit y of Issues	EP5 Extent of Applicabl e Codes	EP6 Extent Of Stake- holder Involveme nt	EP7 Interdependen ce
✓		✓				✓

EP1- Knowledge Dept: It's not reasonable to expect basic engineering principles alone to be enough for developing a successful machine learning-based pneumonia detection system with Explainable AI (XAI). This project requires knowledge of medical imaging concepts, deep learning models and clinical diagnostic constraints. A sound understanding of convolutional neural networks, transfer learning (DenseNet121), image preprocessing methods and model optimization methods is required for validation. Further, incorporating Grad-CAM for interpretability requires knowledge of feature visualization method and its application in clinical decision support

EP3 – Depth of Analysis: The planned system demands the deep and multi level logic based reasoning to accurately diagnose pneumonia from chest X-ray images. The model needs to process intricate and subtle patterns of lung region, capture hierarchies pattern from DenseNet121 architecture, and distinguish normal cases from pneumonia in the presence of an imbalanced data. The framework isn't limited just to classification but also provides performance analysis with the clinical parameters -Recall, F1-Score essential to reduce miss on TB cases and it includes clinically important metrics such as recall and F1-score. In addition, Grad-CAM analysis are conducted to explain model predictions and ensure that the attention is paid on medically informative lung regions. This cluster of feature extraction, performance evaluation, and explainability shows a very deep and complex depth of analysis which fits into the methodics of complex engineering problem-solving.

EP7 – Interdependence: The suggested system of pneumonia detection has high levels of interdependence between the elements. The performance of the deep learning model directly depends on the quality of the chest X-ray dataset. The management of class imbalance affects the recall and false negatives which are important in medical diagnosis. DenseNet121 architecture selection has an influence on the ability to extract features, which consequently has an influence on the accuracy of Grad-CAM visual explanations. Moreover, explainability determines the degree of trust in doctors and other healthcare professionals, which depends on the reliability of the explainability. Any change in a single aspect of the system, including data preProcessing, model architecture, or explainability method has severe implications on the overall system performance and clinical usability. The interdependence is definitely evident in this high dependency among system components as it meets the EP7 requirement of a complex engineering problem.

Mapping with Knowledge Profile

This subsection from the complexity of the proposed system to the corresponding knowledge areas used to solve EP1 (Depth of Knowledge). The production of the model required both programming skills and knowledge about deep learning fundamentals, medical related

domain expertise, a level of systems design/ theory and engineering best practices, in addition to research comprehension. The selected knowledge profiles demonstrate that a combination of several areas were necessary for achieving EP1.

Table 5.6: Mapping with knowledge Profile.

K1 Natural Science	K2 Mathematics	K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K7 Comprehension	K8 Research Literature
		✓	✓	✓	✓	✓	✓

K3 – Background into the engineering fundamentals: This demonstrates applying fundamental Engineering principles around image processing, neural network architecture, optimisation and computational workflows. In order to develop an efficient pneumonia detection system, it is important to understand the convolutional operations, activation functions and model evaluation metrics.

K4 – Expertise: The reported method relies on expertise in deep learning and medical imaging. Information from DenseNet121, transfer learning and X-ray reading are vital to identify useful features from lung areas in order to develop the model correctly.

K5 -- Engineering Design: The creation of an AI diagnostic tool is designing the whole system starting from the preparation of datasets to deployment of models. Choosing architecture components, incorporating Grad-CAM and balancing performance are structured engineering design options.

K6 – Engineering Practice: That use of real engineering tools and industry standards (TensorFlow, Keras, and Google Colab GPUs) to deliver a clear solution. The utilization of the class imbalance treatment, fine-tuning, and performance verification mirrors hands-on engineering work.

K7 – Understanding: A comprehensive understanding of pneumonia pathology, imaging features and deep learning behavior is required to interpret model performance and examine Grad-CAM visualization. You need to understand not only the coding but also why does it act so.

K8 – Combining literature/research: The project is based on several publications, comparing methods, identifying gaps (e.g. no explainability) and justifying the need of our approach. The methodology and design decisions started from a literature review, an analysis of scholarly literature.

5.4.2 Engineering Activities

This section describes how the development of the pneumonia detection system can be

conceptualized through engineering activity categories. The system is a multi-layered one, of technical choice and cost, innovation and effect on society and the mapping enables us to comprehend the scale and complexity of the engineering effort involved.

Mapping with Complex Engineering Activities

Table 5.7: Mapping with Complex Engineering Activities.

EA1 Range of re- sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓	✓	✓	

EA1-Evidence of resources: Diverse computational resources including power systems for GPU training, medical imaging data sets, deep learning frameworks and XAI platforms suggest that the resource aspects is wide and complex.

EA3-Innovation: DenseNet121 with Grad-CAM integration for pneumonia diagnosis presents an innovative way of not only detecting the disease but explaining the prediction regions which increases clinical trust better than conventional AI models.

EA4- Consequences for society and environment:The impact of the system on society is clear in that it supports early detection of pneumonia, and shortens time to diagnosis, potentially saving lives.

5.5 Summary

This chapter reviewed engineering standards, design constraints and challenges of implementing the intended Pneumonia targeted systemUnderTest. We verified relevant software, hardware and communication standards in order to guarantee that the system is reliable, scalable, and fulfills needs of medical data processing. The chapter illustrated how CUDA-enabled GPU platforms, IEEE documentation standards, and JSON communication forms contribute to an efficient model training, testing and deployment. The chapter underscored the societal and ethical consequences of AI in healthcare, including better diagnostic accuracy, reduced clinical burden, and responsible management of personal patient data. A financial analysis of student and commercial implementation cost compared the costs of implementation at such large scale to that of a public microgrid, providing substance for the project's viability and real-world impact.

Finally, the project was aligned with the complexity of engineering problems, knowledge profile and engineering activity aspects; it proved to be a task comprising advanced knowledge area, innovative practice, social implication involvement and resource use. In summary, the chapter shows that the system has not just met technical specifications but also fulfills moral, economical and professional engineering standards and requirements for prospect use in health care environments.

Chapter 6

Conclusion

In this chapter, I summarize the contribution of work in the project. It reiterates the aims, highlights salient findings and concludes by discussing the contributions made toward the construction of the pneumonia detection model.

6.1 Summary

Overall, this chapter focused on the project's successes whole-heartedly proving that the designed pneumonia-detection system has fully satisfied its objectives. The model performed well in detecting the pneumonia from chest X-ray images having high recall and F1-scores which are crucial to medical diagnostics. Incorporating Grad-CAM further improved the system by generating visual justifications for its decisions, and helped to render the results more interpretable and reliable for clinical use. This section establishes that the project makes a significant contribution to AI-assisted healthcare and provides a solid platform on which improvements can be made as well as new functionality added.

6.2 Limitation

While the proposed pneumonia detection system showed promising theoretical results, we uncovered several limitations in this study. First, the model was trained with a public accessible dataset, which may fail to capture all patient conditions, imaging devices or demographic diversity in real clinical practice. Therefore, its performance could be inconsistent between X-ray images from different hospitals or scanning equipments. A second limitation is that the system is binary, classifying only Normal cases from Pneumonia ones. It does not distinguish among the different subtypes of pneumonia (such as bacterial, viral or COVID-related) or identify other types of lung pathology with a similar appearance on imaging. This limits the diagnostic range of evtCP compared with what clinicians encounter in the practice. Furthermore, the model is based on deep learning, which can demand lots of computation resources when trained. Inference with the trained model is feasible for deployment, however need for GPU systems might prevent some institutions from early experimentation. One limitation of the Grad-CAM explainability module, despite being useful, is that it only show important regions of the image but does not convey complete clinical rational which render expert's interpretation mandatory.

Finally, the system as implemented does not incorporate user authentication/login security, storage of data securely or integration with patient management systems which are essential to deploy in end-to-end healthcare networking. These limitations suggest areas for future improvements to make the system more robust, clinically comprehensive and applicable for real world usage.

6.3 Future Work

While the developed pneumonia detection system worked well, some future research and development that may increase clinical applicability and technical robustness of could be drawn forward. One such natural extension is generalizing the system beyond binary classification. Subsequently, novel models can be trained not only to distinguish between Normal and Pneumonia cases, but also among different pneumonia types and also identify other types of lung diseases like TB, fibrosis or pleural effusion. It would increase the system's comprehensiveness and make it much more applicable to real clinical environments. A further direction worth exploring is to introduce more massive and diverse datasets from various hospitals and geographic areas. It would help reduce dataset bias and promote the model's generalization on real-world X-ray images. Future work might also consider employing more advanced architectures, e.g., vision transformers or combined CNN-attention models, for possibly even better function of feature extraction and diagnosis. Consideration Implementation-wise, simple web or mobile based application will enable healthcare workers, particularly of rural/under-resourced areas to easily access the system. For a secure system the ability to securely encrypt data, manage patient details and provide user authentication functionality would also be required to meet medical data protection requirements.

Lastly, additional work can further develop the explainability aspect by integrating more sophisticated XAI methods that provide for more comprehensive and clinically interpretable explanations. Validating and refining these visual explanations with radiologists would further establish the clinical validity of the system. Altogether, such enhancements would provide a pathway to evolve the prototype into a fully deployable medical support tool that can aid clinicians in performing real time diagnostic tasks.

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